

Final Exam Review

Note: The problems in this review are to give students an idea of question that can be asked based on the topics covered. Students are responsible for studying all other materials as not every single topic is covered in the review document.

1. Find a context-free grammar $L = \{a^n b^m c^k : k = |m - n| \text{ and } l, m, n \geq 0\}$
2. Find an s-grammar for the language $L = \{a^n b^n \mid n > 0\}$.
3. Show that the following grammar is ambiguous.

$$\begin{aligned} S &\rightarrow AB \mid aaB \\ A &\rightarrow a \mid Aa \\ B &\rightarrow b \end{aligned}$$

4. Remove λ -productions from the following grammar G .

$$\begin{aligned} S &\rightarrow Aa \mid bBb \\ A &\rightarrow bbB \mid S \\ B &\rightarrow C \mid aa \\ C &\rightarrow bB \mid \lambda \end{aligned}$$

5. Remove unit-productions from the following grammar G .

$$\begin{aligned} S &\rightarrow Aa \mid bBb \\ A &\rightarrow bbB \mid S \\ B &\rightarrow C \mid aa \\ C &\rightarrow bB \mid A \end{aligned}$$

6. Find an NPDA with 2 states that accepts the following language based on the regular expression $r = (aa)^+(ba)^*c$.
7. Let x be a positive integer represented in unary, $x > 0$. Design a Turing Machine (using a transitional graph) that computes the function:

$$f(x) = \begin{cases} x - 1, & \text{if } x \geq 3 \\ x + 3, & \text{if } 0 < x < 3 \end{cases}$$

Computation

$$q_0 w \vdash^* q_f f(w)$$

8. Let $BAL_{DFA} = \{\langle M \rangle \mid M \text{ is a DFA that accepts some string containing an equal number of 0s and 1s}\}$. Show that BAL_{DFA} is decidable.
9. Show that A is decidable iff $A \leq_m 0^*1^*$.
10. Let

$$MODEXP = \{\langle a, b, c, p \rangle \mid a, b, c, \text{ and } p \text{ are positive binary integers such that } a^b \equiv c \pmod{p}\}$$
 Show that $MODEXP \in P$.
11. True or False: The first problem to be classified as NP-Complete was 3SAT.
12. True or False: The Cook-Levin Theorem states that SAT is NP-Complete.
13. Design a Turing Machine (using a transitional graph) that erases the right half of an input string of 1's. For example, if the input string contains 4 1's (1111), then the output should be 2 1's (11). Note, if the original input size is odd, then take the ceiling. For example, if the input string contains 5 1's (11111), then output should be 3 1's (111).

Computation

$$q_0 w \stackrel{*}{\vdash} q_f f(w)$$